OPTI-DRY

OPERATING MANUAL



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Heritage & Innovation
in Textiles





OPTI-DRY (MK4) OPERATING INSTRUCTIONS

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1.0 UNPACKING AND ASSEMBLY

When packed for export, the machine may either be in a wooden packing case (1) or a heavy-duty cardboard container on a pallet (2).

1) If wooden packaging, the top of the packing case should be removed first. A screw driver with the correct bit should be used to remove the lid and any packing batons. Preserve the packing materials should they be needed again.

2) If packed on a heavy-duty cardboard container, first remove the strapping which holds the machine onto the wooden pallet. Slide the cardboard container away from the machine then lift the machine from the wooden pallet.

The machine can be lifted from the packing case or pallet.
TAKE CARE WHEN MANUALLY LIFTING THE MACHINE (Nett Weight ~31 kgs).

Inspect the machine and report any damage. If there are any additional accessories, check that these are complete. Retain the packing materials for use in the event of shipping on or returning the machine to Roaches International Ltd. In this event goods, returned or shipped on, are the responsibility of the sender and the sender is responsible for ensuring that the goods are suitably packed to prevent damage in transit.

2.0 SUMMARY

Engineered for the demands of the modern textile laboratory, the OPTI-DRY precision tumble dryer is quickly becoming the go to standard reference drying unit for textile samples. Leading chain stores and their suppliers are increasingly adopting the OPTI-DRY, recognizing its precision and reliability.

2.1 RANGE OF TESTS

- ISO 6330 – Domestic Washing and Drying Procedures for Textile Testing.
- AATCC LP1 – Laboratory Procedure for Home Laundering.
- Marks & Spencer P1A – Wascator Stability Washes (Tumble Dryer).

Please refer to technical datasheet for applicable standards.

The goods mentioned conform but is not limited to the mentioned standards. For more standards conformance please email sales@roaches.co.uk or call +44 (0)1924 420083.

3.0 SAFETY

The OPTI-DRY incorporates a safety switch to prevent rotation and heating if the door is opened. A thermal cut-out to prevent overheating protects the process. The temperature controller should not be set to a heat above 80ºC (176ºF).

The metal drum will become hot during operation and if the normal cooling cycle is not performed will remain hot for some time. Exercise caution if the drum is still hot when loading or unloading test specimens.

- The machine must not be used with the wrong power supply. Use the voltage and frequency which stated on the serial plate. Failure to do so may cause permanent damage to the machine.
- The machine must be supplied with a power source which is grounded properly. This will ensure personal safety and prolonged life of the machine.
- **ALWAYS** wear protective gloves and clothing when checking and handling the machine.
- **DO NOT** wear loose clothes when the machine is in operation.
- **NEVER** load specimens that are dripping wet into the equipment.
- **DO NOT** load specimens that are contaminated with flammable substances.
- **DO NOT** direct the exhaust air towards the air intake on the equipment.
- **DO NOT** overload the equipment (maximum load is 8 kg dry weight).
- **DO NOT** turn the equipment off with a hot load in the drum.
- **DO NOT** add liquids to the specimens in the drum.

4.0 INSTALLATION

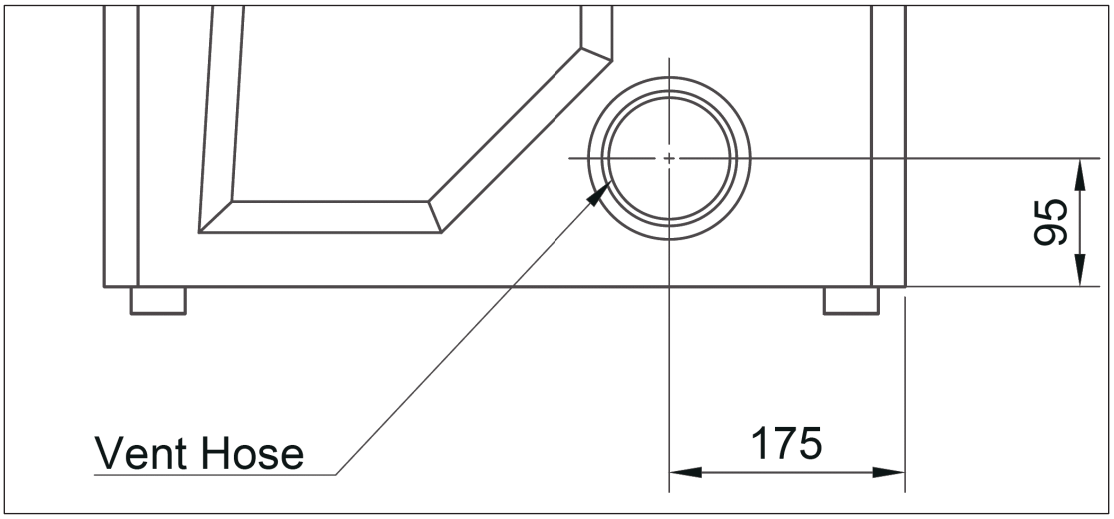
4.1 SITTING

The dimensions of the OPTI-DRY are 595 x 587 x 846mm and can therefore be accommodated with standard work top installations but special attention should be given to the adequate ventilation of the equipment.

If the equipment is to be installed underneath a worktop, allow an air gap above the equipment of no less than 10mm. If the equipment is to stand on a carpeted floor, ensure that the equipment has an air gap between the carpet and the base plate. If the base plate is restricted, the equipment may overheat. Ensure that the equipment is level.

4.2 EXHAUST

Access is only required from the front of the equipment in order to operate it but consideration should be given to the exhaust outlet which should be vented outside to prevent condensation problems in the laboratory. The OPTI-DRY vents from the rear of the equipment, care should be taken not to obstruct the exhaust outlet point. The length of the hose must not exceed 2.4m.



4.3 ELECTRICAL CONNECTION

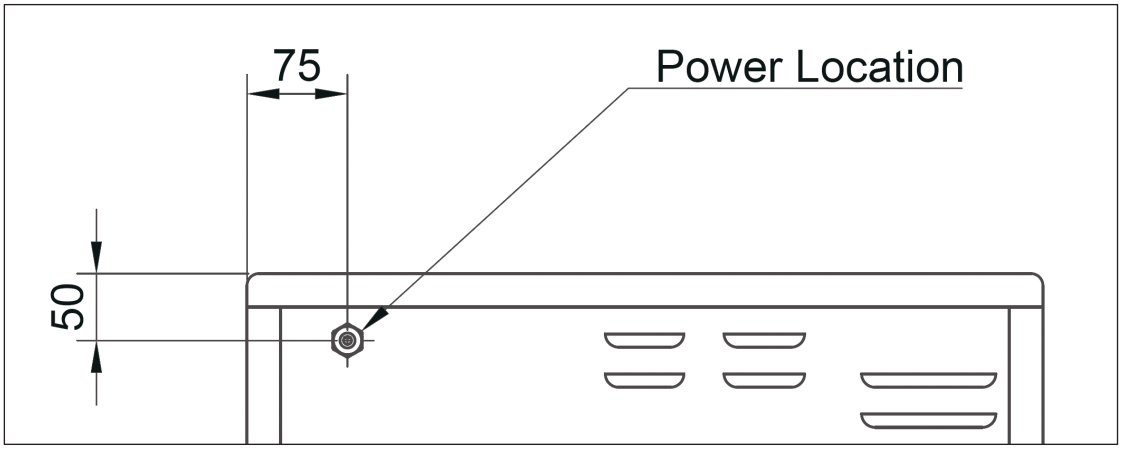
The OPTI-DRY unit is supplied as standard with a Type G UK Mains Plug (BS 1363). Seek professional help when changing this to a plug of a different country or type.



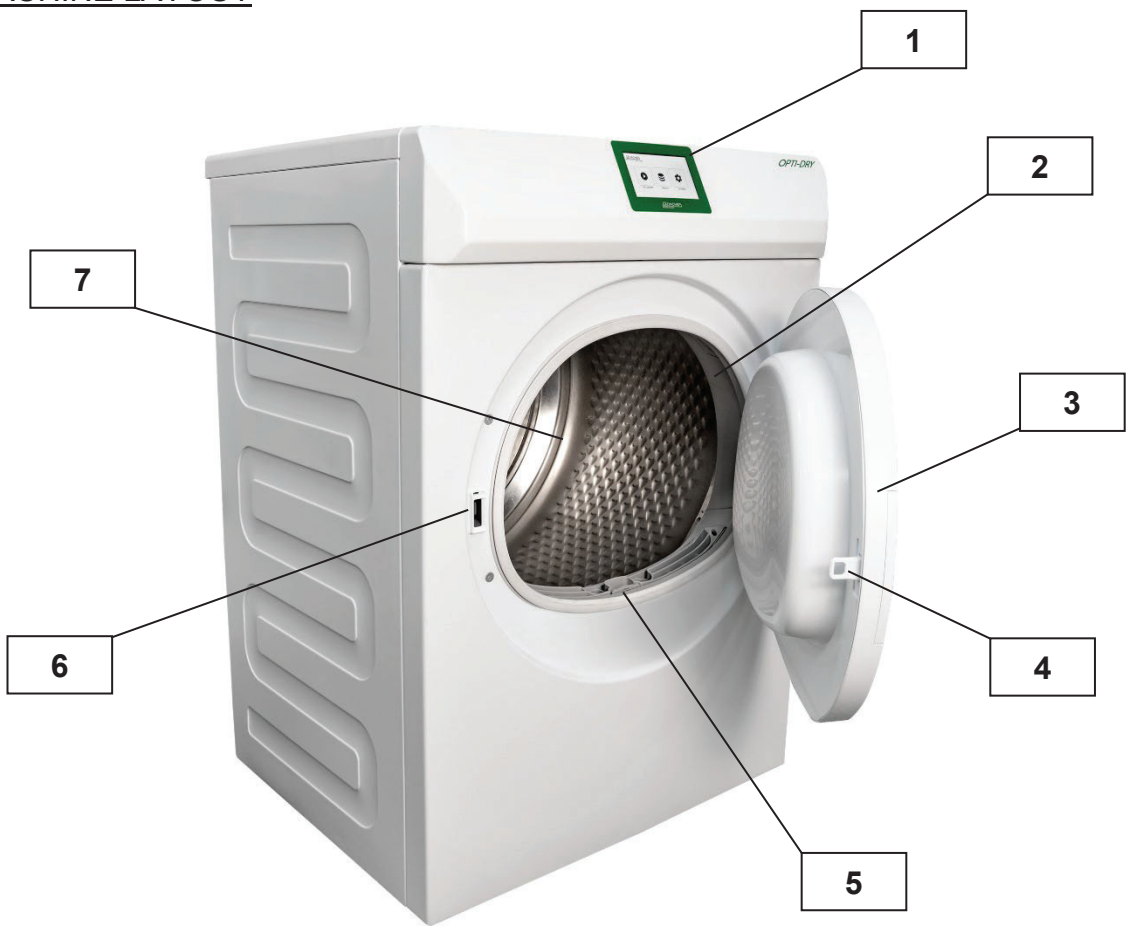
THIS EQUIPMENT MUST BE EARTHED

Note that the mains lead wires are coloured in accordance with the following code

Green/Yellow	**=**	**Earth**
Blue	**=**	**Neutral**
Brown	**=**	**Live**



5.0 MACHINE LAYOUT



1. Touchscreen Controller
2. Door Seal
3. Drum Door
4. Door Catch
5. Lint Filter
6. Door Interlock
7. Drum

5.1 PARAMETERS

Power Supply:	220V to 240V, 50 or 60HZ (Single Phase)
Amp:	13A
Wattage:	3,250W
Weight:	31kg
Dimensions :	595mm (L) x 587mm (D) x 846mm (H)

6.0 OPERATION

6.1 LOADING THE DRUM

The maximum dry load that can be accommodated is 8kg.

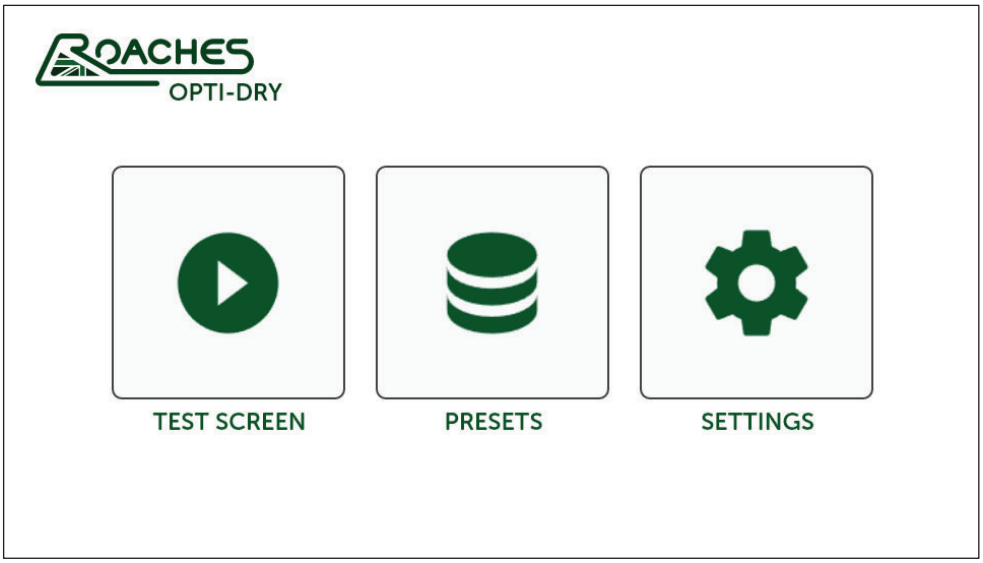
DO NOT OVERLOAD THE DRUM.

When loading the specimens, push them to the rear of the drum to avoid trapping in the door seal when the door is closed.

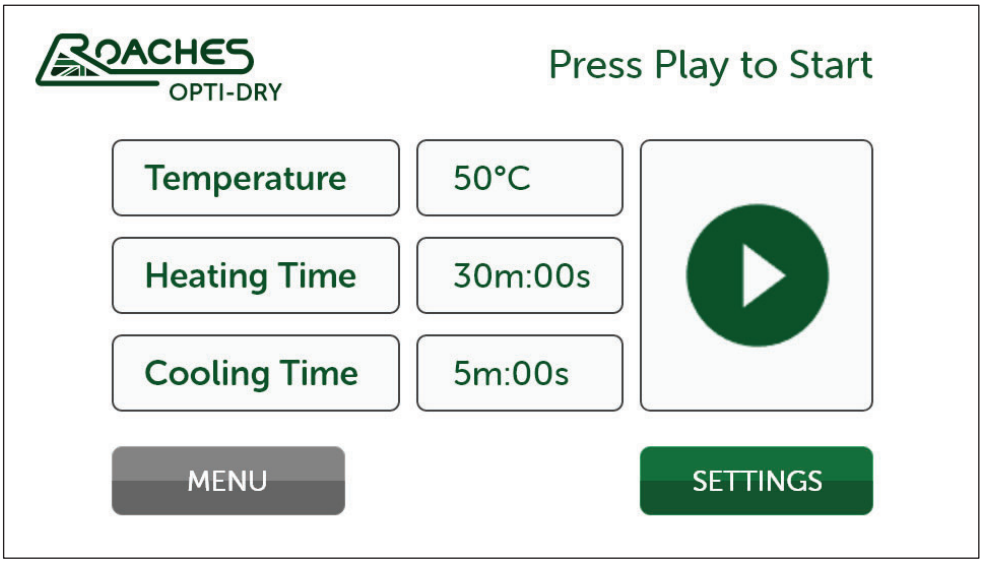
6.2 TOUCHSREEN CONTROLLER: MENUS



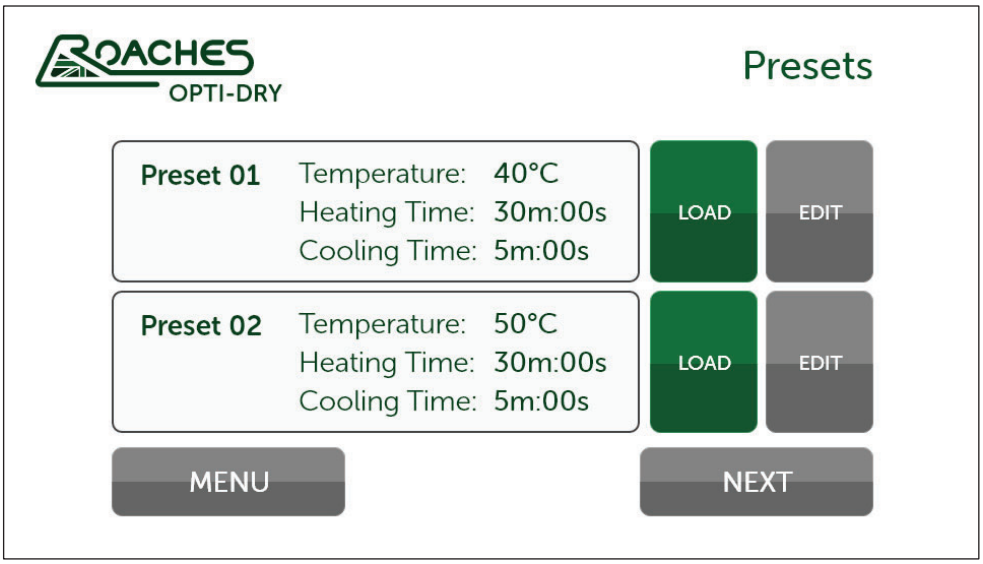
1) Boot Screen



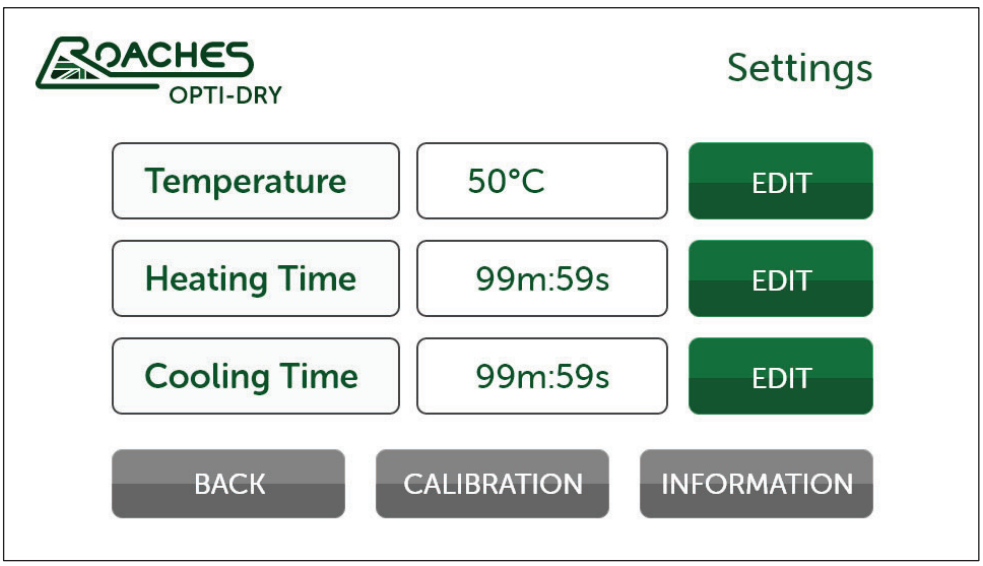
2) Main Menu



3) Test Screen

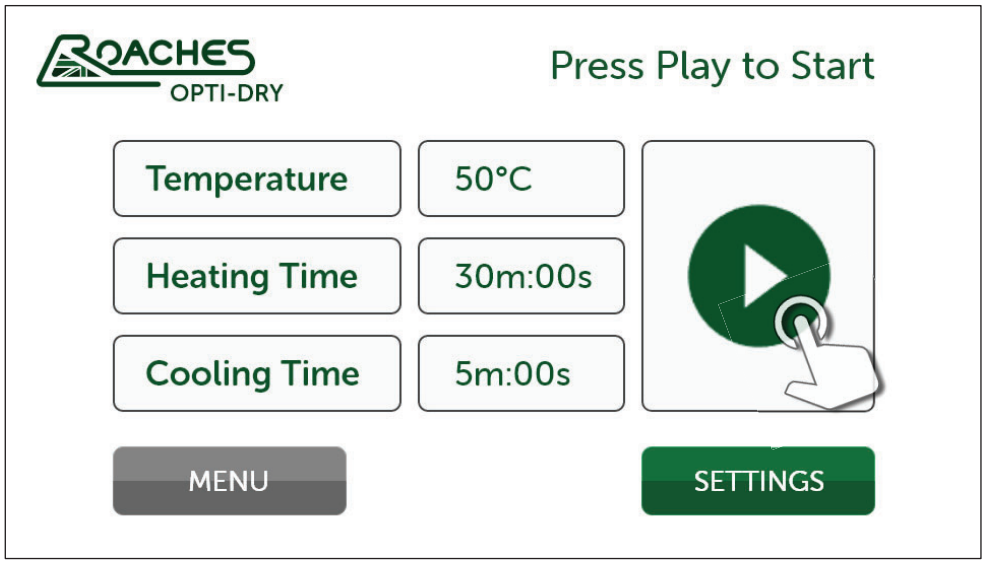


4) Presets



5) Settings

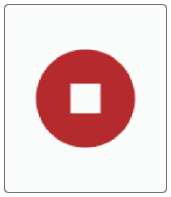
6.3 TOUCHSREEN CONTROLLER: TEST SCREEN





PLAY / RESUME

This button will start the selected program or continue the program is stopped for any reason.



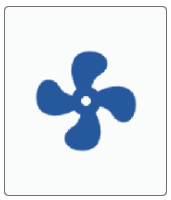
STOP / END

Pressing this button will terminate the current program.



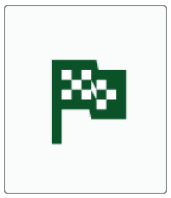
HEATING

This icon will be displayed when the machine is in heat mode. The icon will glow when the heater is on.



COOLING

This icon will be displayed when the machine is in cooling mode.

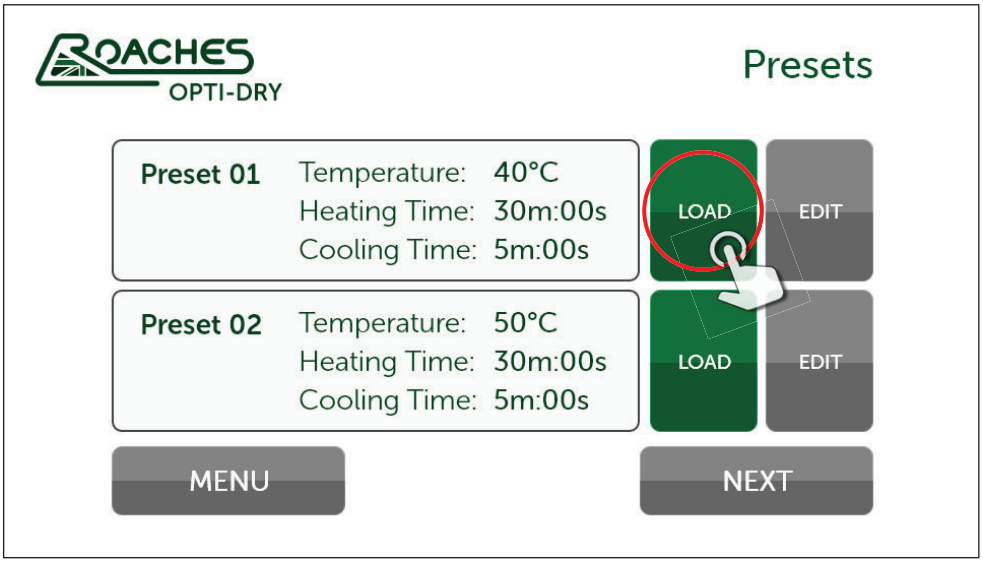


FINISHED

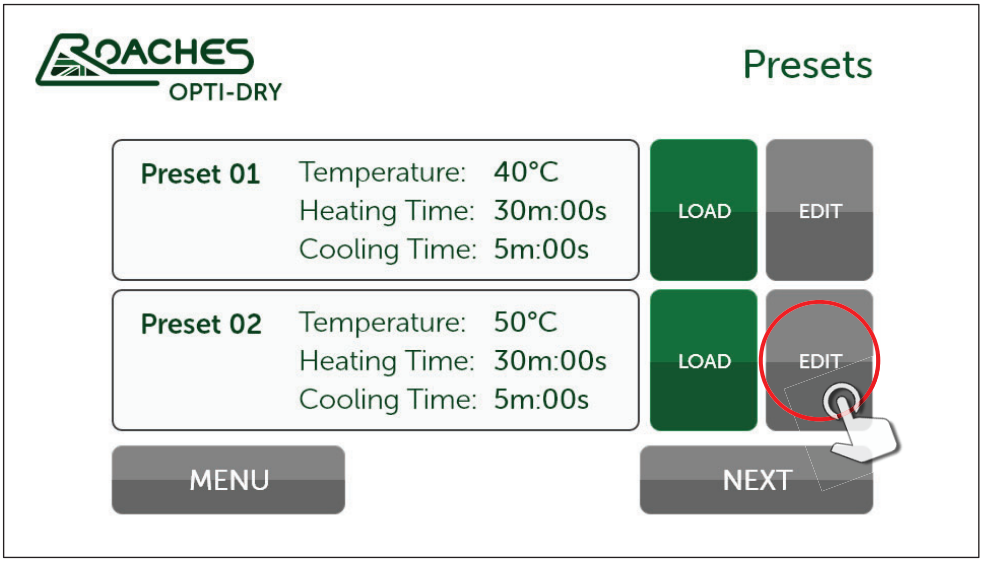
This icon will be displayed when the selected program has finished.

6.4 TOUCHSREEN CONTROLLER: PRESETS

The OPTI-DRY is capable of storing 6 presets which can be loaded direct from the presets page.



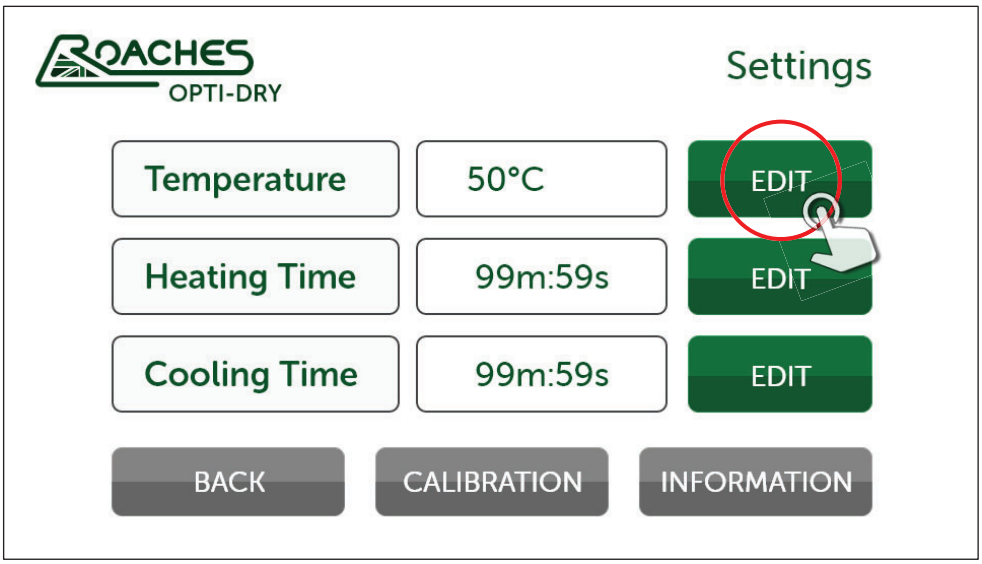
To transfer the settings to the test screen, find the preset required and press the 'LOAD' button.



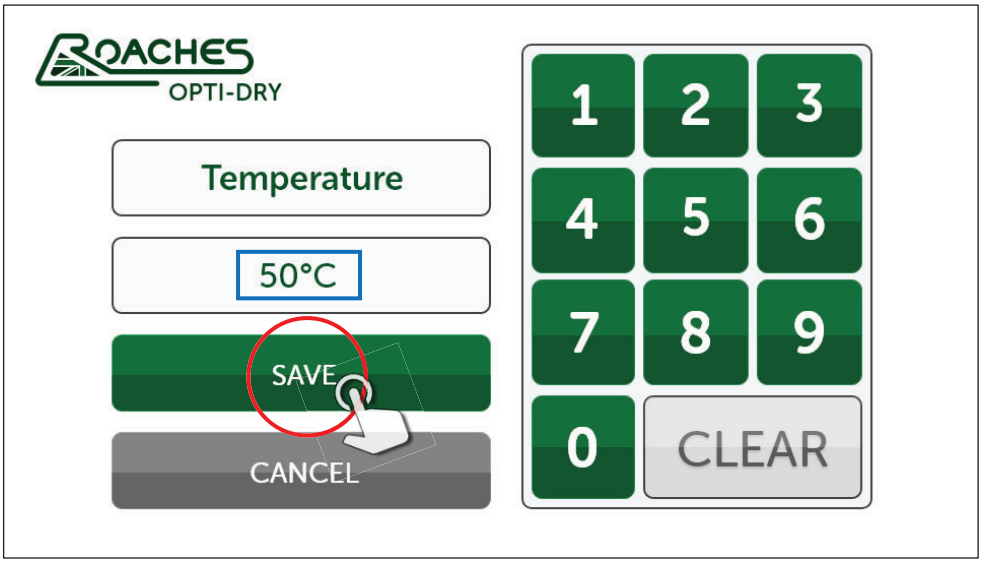
Presets can be modified by pressing the edit button.

6.5 TOUCHSREEN CONTROLLER: SETTINGS

The settings page is where you can edit the current ‘Temperature’, ‘Heating Time’ or ‘Cooling Time’. You can also view the machine information.
The ‘CALIBRATION’ page is for Roaches engineers only.
The ‘CALIBRATION’ page is password protected.



Press the ‘EDIT’ button to change the current settings (see next step below).



Enter your required parameter by pressing the figure marked blue. Use the keypad to enter the parameter and finally press ‘SAVE’ to make the change.

7.0 MAINTENANCE

The OPTI-DRY has few moving parts and therefore maintenance is limited to the following simple tasks. These items are important however and should not be neglected because of their effects on the performance of the process.

Access to the working parts is by removal of the screws which are the rear of the machine. (see SAFETY section 3.0). **Always** turn off the power before removing the access panel.

THE FOLLOWING ATTENTION IS ADVISED

DAILY CHECKS

Ensure the equipment is kept clean, particularly the drum and exhaust duct. Wipe out the drum with a cloth to remove all traces of lint and loose threads that inevitably build up during use.

Remove the **Lint Filter** after each cycle and remove the fluff from it and the filter holder before it is replaced. **Never use the equipment without the lint filter.**

MONTHLY CHECKS

Inspect the **Bearing Pads** and **Support Rollers** to the open end of the drum. If the pads or rollers are worn, they must be replaced. Failure to replace may result in damage to the drum or the drum bearings.

Inspect and check the **Drive Belt**. Look for any signs or wear or damage.

The **Drum Bearings** are made from special self-lubricating materials and therefore it is not recommended or necessary for any additional lubrication during their service life.

Check the **Door Seal** for any signs of wear. Damage to the seal could make the machine less efficient and loose heat unnecessarily.

Roaches recommend that the **Lint Filter** is changed every 12 months or when it shows signs of wear or as soon as it is damage.

8.0 REPLACEMENT PARTS

The performance and efficiency of the equipment you are using can only be maintained if the operation and routine servicing program recommended in the instruction manual is adhered to.

Some components will eventually need to be replaced as part of the maintenance program.

Many parts are held in stock for a period of time to provide back up for older equipment, even if the part is obsolete, obsolescent or no longer fitted to the current equivalent machine. Occasionally however, it may not be possible to supply the part that was originally fitted. In this instance an equivalent part will be offered.

Replacement parts should be ordered from an authorised Roaches International appointed agent.

For best results and optimum performance:-

1. Service and maintain the machine in accordance with the recommendations of the manual.

2. Only use parts recommended by Roaches International Ltd.

ORDERING REPLACEMENT PARTS

To reduce delivery times, enter the following information on the purchase order:-

1. **QUANTITY**- State the number of items required.

2. **ITEM DESCRIPTION** - Where possible state the item code number and description.

3. **MACHINE TYPE** - e.g. Pyrotec, Washtec, Durawash etc.

4. **SERIAL No.** This is applied on the ‘data plate’ fixed to the machine back panel and is a five digit number e.g. 20095.

5. **SUPPLY VOLTAGE** - This information is specified on the ‘data plate’.

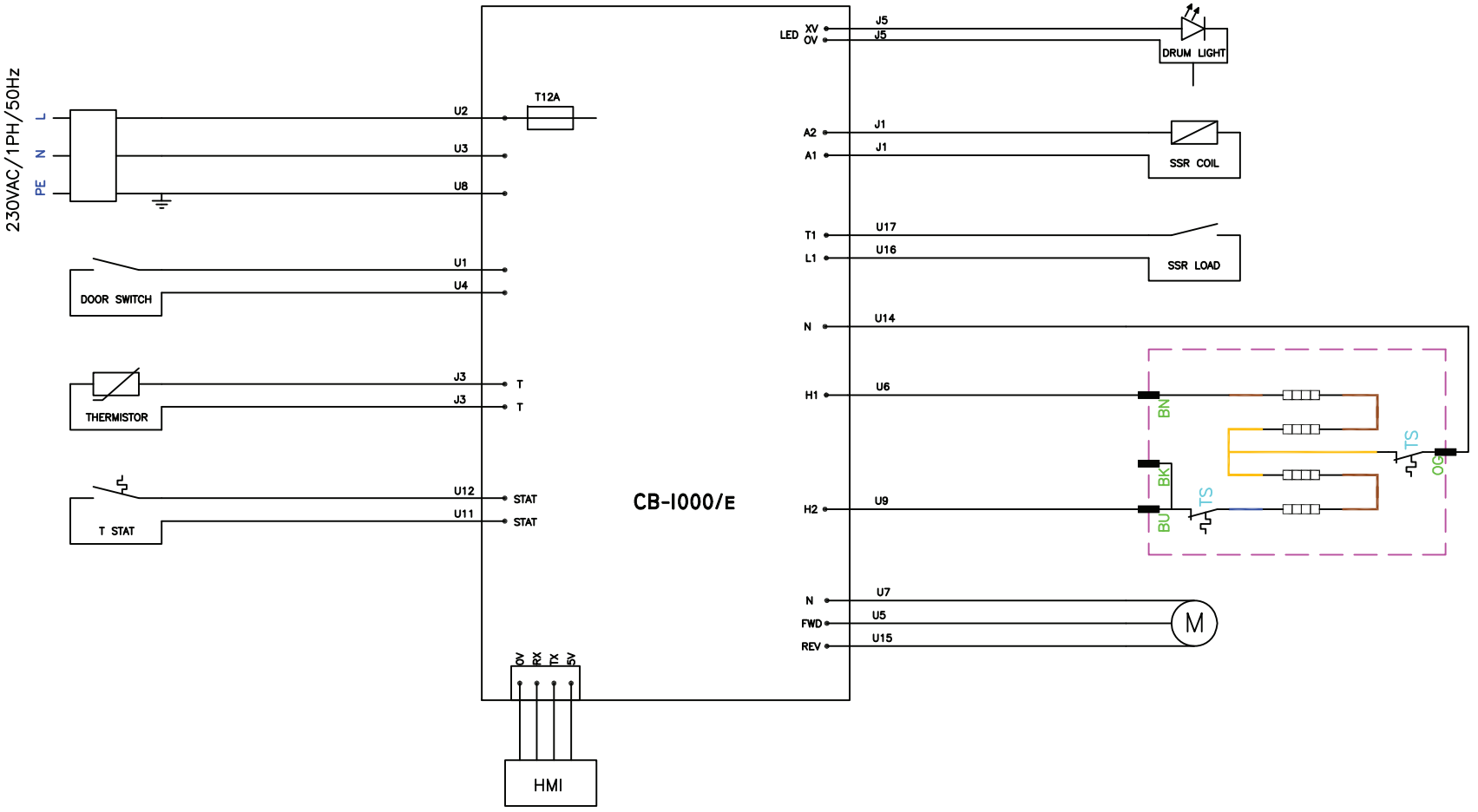
6. **SUPPLY FREQUENCY** - This information is specified on the ‘data plate’.

9.0 AMENDMENT HISTORY

Issue A – Original Issue - July 2025

A4

DO NOT SCALE



REV	C.A.R.	DESCRIPTION	BY	DATE	APP	DATE
1		ORIGINAL ISSUE	MRH	18/07/25	LO	18/07/25

Title
OPTI-DRY
PCB
230V_CIRCUIT

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Drawn	Date	Original Scale	Drawing Number
MRH	17/07/25	N/A	WD1698A